

BISA SAITEJASWINI RAO

Robotics & Automation Engineer

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My Portfolio- https://saitejaswinirao.github.io/

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Career Objective

Motivated and detail-oriented Engineering graduate with hands-on experience in robotics, automation, embedded systems, and software technologies. Passionate about applying technical knowledge, analytical thinking, and problem-solving skills to design, optimize, and develop innovative and efficient solutions. Seeking a challenging opportunity to contribute to organizational success through continuous learning, collaboration, and technical excellence while driving innovation, improving operational efficiency, and delivering high-quality results in a dynamic and growth-oriented environment.

Education

Aug 2025 – Present
Gandhinagar, Gujarat
International Professional Master's Program in Advanced Robotics Technology **8.75**
New Age Makers' Institute of Technology

Feb 2023 – Jun 2025
Guntur, Andhra Pradesh
B.Tech (Honours) in Product Design **8.51**
Vignan's Foundation for Science, Technology & Research

Aug 2022 – Jun 2025
Guntur, Andhra Pradesh
Bachelor of Technology in Robotics & Automation **8.45**
Vignan's Foundation for Science, Technology & Research

May 2019 – Jun 2022
Hyderabad, Telangana
Diploma in Automation & Robotics Engineering **7.78**
Central Institute of Tool Design (MSME)

Mar 2018 – Apr 2019
Ghatkesar, Telangana
Central Board of Secondary Education (Xth) **7.75**
Vignan Prabodhananda Prashanti Niketan

Experience

Dec 2024 – May 2025
Guntur, Andhra Pradesh
Teaching Assistantship
Vignan's University (VFSTR)
- Delivered and supported coursework in **Fundamentals of Robotics, Mobile Robotics, Robotics for Engineers, and Engineering Drawing**, integrating theory with practical learning.
- Conducted **hands-on labs and tutorials** on **robot kinematics & mobile robot concepts**.
- Mentored students in **robot design, simulation, and project development** using **Fusion 360**, including debugging and performance evaluation.

Oct 2023 – Nov 2023
Hyderabad, Telangana (Remote)
RPA Practitioner Intern
Henry Harvin Education (London)
- Automated **10+ routine workflows** using UiPath, boosting process efficiency.
- Identified **multiple automation opportunities** in collaboration with production teams.
- Tested, deployed & monitored **high-reliability bots**, ensuring smooth performance.

Projects

Sep 2025 – Mar 2026
Design & Control of Warehouse Mobile Robot
- LIDAR-based navigation
- Designed a material dispensing unit with a planar manipulator

Oct 2025 – Dec 2025
Development of Teleoperation Controller Module
- A 3D-printed enclosure for housing and protection.
- Three control modes – Unilateral, Bilateral, and Virtual Wall.

Oct 2025 – Dec 2025
Fabrication & Control of Soft Gripper
- Pneumatically actuated
- Closed-loop control using a pressure sensor

Feb 2024 – May 2025
Autonomous Rover with Rocker-Bogie Mechanism
- ROS-based control
- Fabricated using FDM technology (PLA & ABS)

Nov 2023 – Jun 2025
A Semi-Humanoid Robot with Autonomous Capabilities
- Facial recognition through OpenCV
- Arms with 6-DOF

Skills

Programming Languages: C, Python, HTML

Technical: ROS, RPA, CAD Modeling (Fusion 360, Catia), PLC, HMI, OpenCV, Robot System Integraion (Fanuc)

Soft Skills: Communication, Project Management, Adaptability, Initiative, Leadership, Teamwork

Courses

Jul 2024 – Oct 2024	**Metal Additive Manufacturing** NPTEL
Aug 2023 – Oct 2023	**RPA Using UiPath Course** NASSCOM
Jan 2023 – Apr 2023	**The Joy of Computing Using Python** NPTEL

Certificates

SOLIDWORKS Design Associate (CSWA), Dassault Systèmes -2026

Business English Certificate Vantage (BEC) – B2 Level, Cambridge University -2024

Preliminary English Test (PET) - B1 Level, Cambridge University -2023

Certified Robotic Process Automation Practitioner, Henry Harvin -2023

Publications

08/2026	**Effect of Infill Pattern on Tensile Properties of PLA ABS Based Hybrid Composites Reinforced with Natural Fibers**
12/2024	**The Impact of Robots in Agriculture for Enhanced Precision in Farming** IJRES Journal

Patents

06/2026	**Autonomous Localization and Mapping Robot for Delivery Applications**
08/2026	**Autonomous Space Exploring Rover with onboard Manipulator**
09/2026	**MANAV-AI-Enabled Humanoid Robotic Assistant**
09/2026	**VINATA-Design of an AI-Enabled Humanoid Mobile Robot**
09/2026	**Autonomous House Cleaning Robot**

Achievements

Secured 1st Place in Robot System Integration, Telangana State - Indiaskills 2025 State Competition

Accomplished 'C' certificate in "National Cadet Corps (NCC)"

Cleared "Quals Round" in IRoCU – 2024

Received "Student SEED Grant" of Rs. 1.9 Lakhs by Vignan's University

Secured "AIR 2" in Janatics Skill Automation Competition

Languages

English | Hindi | Telugu | Marathi